Worksheet # 5 Part B Upload solutions
Physics II

Name: Key Date: _____

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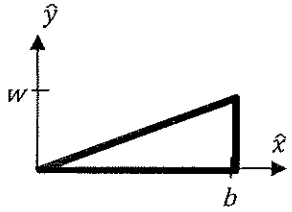
Upload the solution to questions 11 through 15 below

11) A wire triangle with the lengths of the sides b and w sits in a magnetic field pointing in the + $\hat{z}$ direction. The magnetic field has an x-axis and time dependence. Determine the equation for the induced electric potential in the wire triangle given by the equation below where C is a constant.

$\vec{B} = (Cxt^2)\hat{z}$

See attached

$V_{emf} = -\frac{2Cwb^2t}{3}$



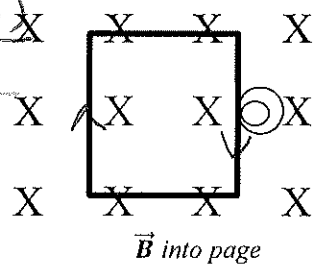
12) A simple square circuit consisting of a light bulb and a rectangular loop of wire (N = 10) with each side having a length of 20 cm, the circuit's area is perpendicular to the magnetic field lines.

A) For the first configuration the magnitude field strength is at 0.5 T and is reduced to 0.1 T in t = 0.01 s, determine the amount and direction of the maximum induced current through the light bulb if the resistance of the light bulb is 50 Ω?

$V_{emf} = -N\frac{d\Phi}{dt} = -N\frac{(B_f A_f \cos\theta_f - B_i A_i \cos\theta_i)}{\Delta t}$

$V_{emf} = -N\frac{(A(B_f - B_i))}{\Delta t} = 16V$

$I = \frac{V_{emf}}{R} = 0.32\ A\ CW$

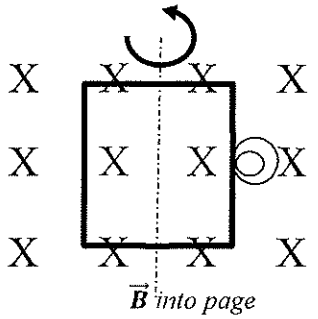


B) For the second configuration the magnitude field strength is held constat at 0.5 T s, the square wire is rotated about its center (dotted line) though 90° in t = 0.01 s so that the light bulb goes into the page and the opposite side from the light bulb comes out of the page. Determine maximum induced current through the light bulb if the resistance of the light bulb is 50 Ω?

$V_{emf} = -N\frac{\Delta\Phi}{\Delta t} = \frac{AB(\cos(\theta_f) - \cos(\theta_i))}{\Delta t}$

$V_{emf} = 2.0V$

$I = \frac{V_{emf}}{R} = 0.4A$



13) A metal detector uses a coil in the shape of a solenoid, with N = 50 loops in the coil, a cross sectional area of A = 0.002 m^2 and a length of l = 5.0 x 10^{-3} m. The solenoid uses an a.c. current running through the coils that is changing at 0.03 A in 2 x 10^{-3} s. If the inside of the solenoid goes from filled with Cobalt (μ_{cobalt} = 3.1 x 10^{-4} Tm/A), to filled with Iron (μ_{iron} = 6.4 x 10^{-3} Tm/A), what is the change in the induced voltage across the solenoid between the two metals?

$L_C = \frac{\mu_C N^2 A}{l} = 0.31\,H$

$L_I = \frac{\mu_I N^2 A}{l} = 6.4\,H$

$V_C = -L_C\left(\frac{\Delta I}{\Delta t}\right) = 4.65\,V$

$V_L = -L_I\left(\frac{\Delta I}{\Delta t}\right) = 96\,V$

$\Delta V = 91.35\,V$

14) An inductor is wired in series with a open switch and a dc power supply. If when the switch is closed the current changes are given by the equation below. Determine the self-inductance L if when the switch is closed after t = 0.01 s the induced voltage across the inductor is 4.76 V. The constant a = 5/s and I_o = 20 A.

$$I(t) = I_0(1 - e^{-at})$$

See attached

$L = 0.05\,H$

15) A RCL series circuit has a capacitor (C = 20 μF), inductor (L = 2 mH) and a resistor (R = 100 Ω).

A) The ac power supply is set at 1,000 V rms and at a frequency of f = 5 kHz, determine the current through the circuit.

$X_C = \frac{1}{2\pi fC} = 1.6\,\Omega$

$X_L = 2\pi fL = 62.8\,\Omega$

$R = 100\,\Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2} = 117.2\,\Omega$

$V = IZ$

$I = \frac{V}{Z} = 8.5\,A$

B) Determine the resonance frequency for this circuit.

$f = \frac{1}{2\pi\sqrt{LC}} = 796\,Hz$

C) If the power supply is set to the resonance frequency and 1,000 V, determine the volage across the resistor at resonance.

Z at resonance $Z = R$

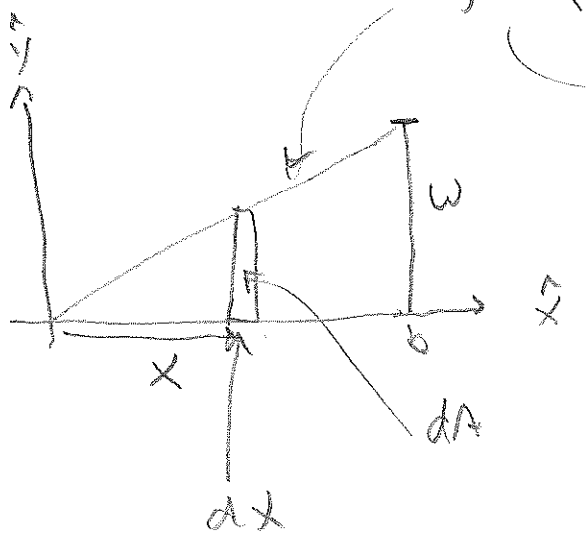
$V = IZ = IR$

$I = \frac{V}{R} = 10\,A$

$V_R = IR = 1{,}000\,V$

11)

$$y = \left(\frac{w}{b}\right)x + 0$$



$$dA = y\,dx = \left[\left(\frac{w}{b}\right)x\right]dx$$

$\vec{B}\cdot\hat{n} = (B)(1)$

$\cos(\phi) = 1$

$\vec{B}$ and $\hat{n}$ are same direction

$$\Phi_m = \int \vec{B}\cdot\hat{n}\,dA$$

$$\Phi_m = \int_0^b \left[(cxt^2)\left(\frac{w}{b}\right)x\right]dx = \frac{ct^2w}{b}\int_0^b x^2\,dx$$

$$\Phi_m = \frac{ct^2w}{3b}\left[x^3\right]_0^b = \frac{ct^2wb^2}{3}$$

$$V_{ent} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\left[\frac{ct^2wb^2}{3}\right]$$

$$V_{ent} = -\frac{2cwb^2t}{3}$$

14) $V_{emf} = -L\frac{dI}{dt} = -L\left(\frac{d}{dt}(I_0 - Ie^{-at})\right)$

$V_{emf} = -L\left(\frac{dI_0}{dt}^{0} - I_0\frac{de^{-at}}{dt}\right)$

$V_{emf} = -L\left(-\left(-aIe^{-at}\right)\right)$

$V_{emf} = -LaIe^{-at}$

$L = -\frac{V_{emf}}{(aIe^{-at})} = -\frac{4.76V}{(5/s)(20A)(0.95)}$

$L = 0.05\ H$